https://arxiv.org/abs/2006.08289

```
================================================
2×2 iPEPS: D=2  χ=20  h=3.000
================================================

── Simple Update ──
  dτ=0.1000 ×20: S=0.0342
  dτ=0.0100 ×40: S=0.0005
  dτ=0.0010 ×60: S=0.0000

── Initial CTMRG ──
  CTMRG   2: ΔSV=6.19e-01
  CTMRG   3: ΔSV=5.23e-04
  CTMRG   4: ΔSV=1.02e-04
  CTMRG   5: ΔSV=3.43e-05
  converged

── Final CTMRG ──
  CTMRG   2: ΔSV=2.89e-04
  CTMRG   3: ΔSV=1.76e-04
  CTMRG   4: ΔSV=1.96e-04
  CTMRG   5: ΔSV=1.26e-05
  converged

── Observables ──
  E/site = -2.7840964972
  ⟨σᶻ⟩   = -0.575768
  ⟨σˣ⟩   = 0.817599
  S_ent  = 0.000007
  E_QMC  = -3.2802  (reference)
Energy = -2.784096
```

```
...
    ================================================
    2×2 iPEPS: D=2  χ=100  h=3.000
    ================================================

    ── Simple Update ──
      dτ=0.1000 ×30: S=0.0491
      dτ=0.0100 ×60: S=0.0010
      dτ=0.0010 ×120: S=0.0000

    ── Initial CTMRG ──
      CTMRG   2: ΔSV=1.12e+00
      CTMRG   3: ΔSV=1.70e-01
      CTMRG   4: ΔSV=1.88e-05
      converged

    ── Final CTMRG ──
      CTMRG   2: ΔSV=7.16e-05
      converged

    ── Observables ──
      E/site = -2.9935706992
      ⟨σᶻ⟩   = -0.112662
      ⟨σˣ⟩   = 0.993629
      S_ent  = 0.000014
      E_QMC  = -3.2802  (reference)
    D=2  E=-2.993571

    ================================================
    2×2 iPEPS: D=3  χ=100  h=3.000
    ================================================

    ── Simple Update ──
      dτ=0.1000 ×30: S=0.0394
```

```
[ ]
println("\nQMC reference (h=3.0): E/site = -3.2802")
```

```
D=3  E=-2.804884

================================================
2×2 iPEPS: D=4  χ=100  h=3.000
================================================

── Simple Update ──
  dτ=0.1000 ×30: S=0.0457
  dτ=0.0100 ×60: S=0.0008
  dτ=0.0010 ×120: S=0.0000

── Initial CTMRG ──
  CTMRG   2: ΔSV=9.16e-01
  CTMRG   3: ΔSV=5.21e-03
  CTMRG   4: ΔSV=1.46e-03
  CTMRG   5: ΔSV=4.57e-04
  CTMRG   6: ΔSV=5.89e-04
  CTMRG   7: ΔSV=4.07e-04
  CTMRG   8: ΔSV=7.42e-04
  CTMRG   9: ΔSV=2.41e-04
  CTMRG  10: ΔSV=6.87e-04
  CTMRG  11: ΔSV=1.05e-03
  CTMRG  12: ΔSV=9.06e-04
  CTMRG  13: ΔSV=2.17e-03
  CTMRG  14: ΔSV=1.24e-03
  CTMRG  15: ΔSV=3.61e-04
  CTMRG  16: ΔSV=2.85e-03
  CTMRG  17: ΔSV=3.51e-03
  CTMRG  18: ΔSV=2.75e-03
  CTMRG  19: ΔSV=2.56e-04
  CTMRG  20: ΔSV=3.07e-03
  max_iter reached
```

```
      max_iter reached

  ••• ── Final CTMRG ──
      CTMRG   2: ΔSV=2.68e-03
      CTMRG   3: ΔSV=1.02e-03
      CTMRG   4: ΔSV=3.28e-04
      CTMRG   5: ΔSV=1.48e-03
      CTMRG   6: ΔSV=9.49e-04
      CTMRG   7: ΔSV=9.48e-04
      CTMRG   8: ΔSV=7.28e-04
      CTMRG   9: ΔSV=4.99e-04
      CTMRG  10: ΔSV=1.57e-04
      CTMRG  11: ΔSV=5.95e-04
      CTMRG  12: ΔSV=1.96e-03
      CTMRG  13: ΔSV=3.24e-04
      CTMRG  14: ΔSV=7.17e-04
      CTMRG  15: ΔSV=9.17e-04
      CTMRG  16: ΔSV=2.23e-04
      CTMRG  17: ΔSV=1.21e-03
      CTMRG  18: ΔSV=4.30e-04
      CTMRG  19: ΔSV=5.76e-05
      converged

    ── Observables ──
      E/site = -2.8989407829
      ⟨σᶻ⟩   = -0.419803
      ⟨σˣ⟩   = 0.907618
      S_ent  = 0.000010
      E_QMC  = -3.2802  (reference)
    D=4  E=-2.898941

    QMC reference (h=3.0): E/site = -3.2802
```

```
  converged

── Full Update (1 sweeps, r_max=4) ──
  ▶ sweep 1/1
  FU A–B
    AB ALS 1: E=-2.85708162
    AB ALS 2: E=-2.85753704
  FU C–D
    CD ALS 1: E=-2.85545811
    CD ALS 2: E=-2.85544202
  FU A–C
    AC ALS 1: E=-2.85547861
    AC ALS 2: E=-2.85552229
  FU B–D
    BD ALS 1: E=-2.85563392
    BD ALS 2: E=-2.85564077
  post-sweep CTMRG...
  CTMRG   2: ΔSV=4.30e-05
  converged
  sweep E=-2.8556959587

── Final CTMRG ──
  CTMRG   2: ΔSV=1.12e-05
  converged
```

```
...  ======================================================
     2×2 iPEPS CONVERGENCE — h=3.00  n_fu=1
     ======================================================

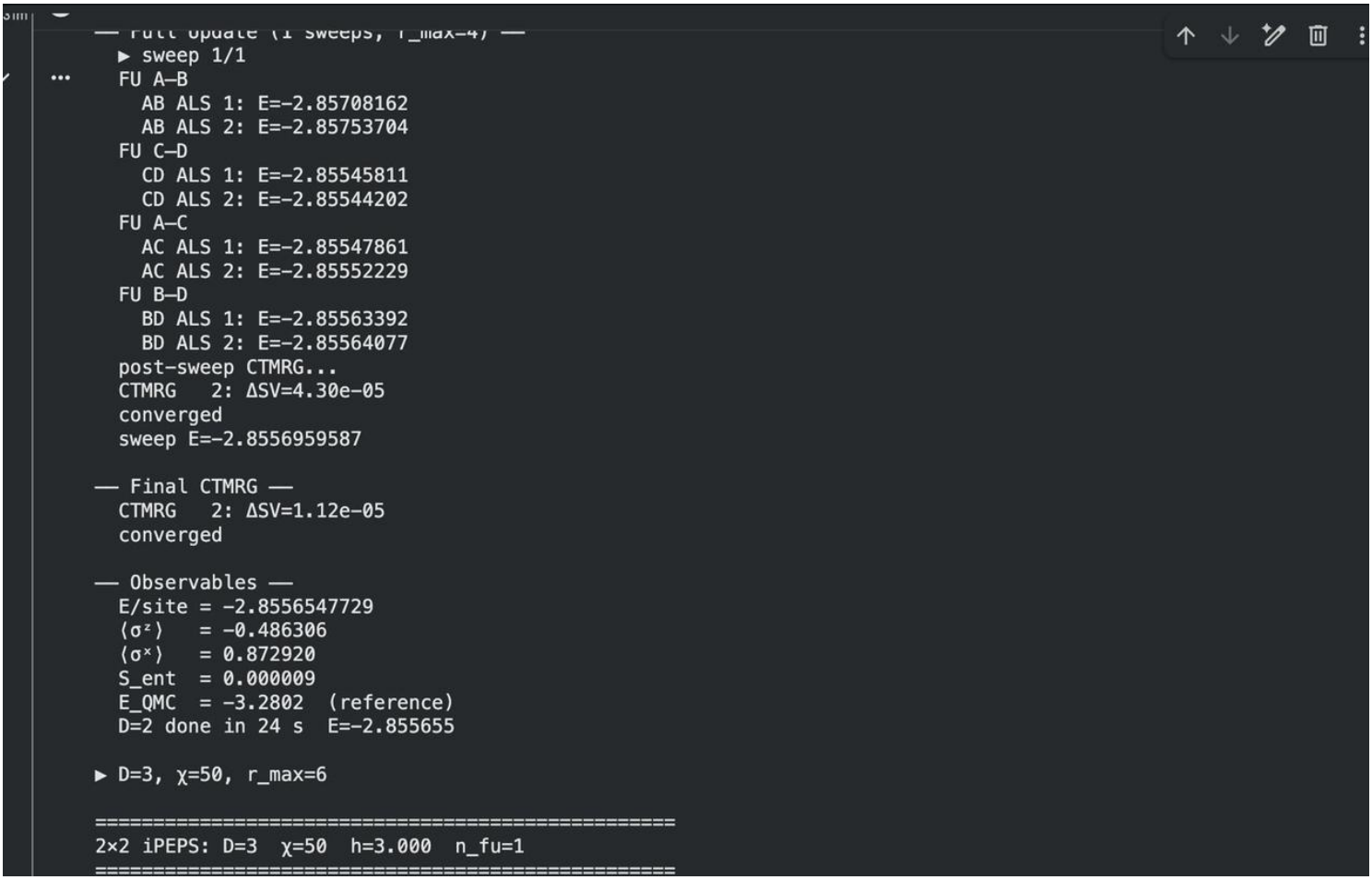
     ▶ D=2, χ=40, r_max=4

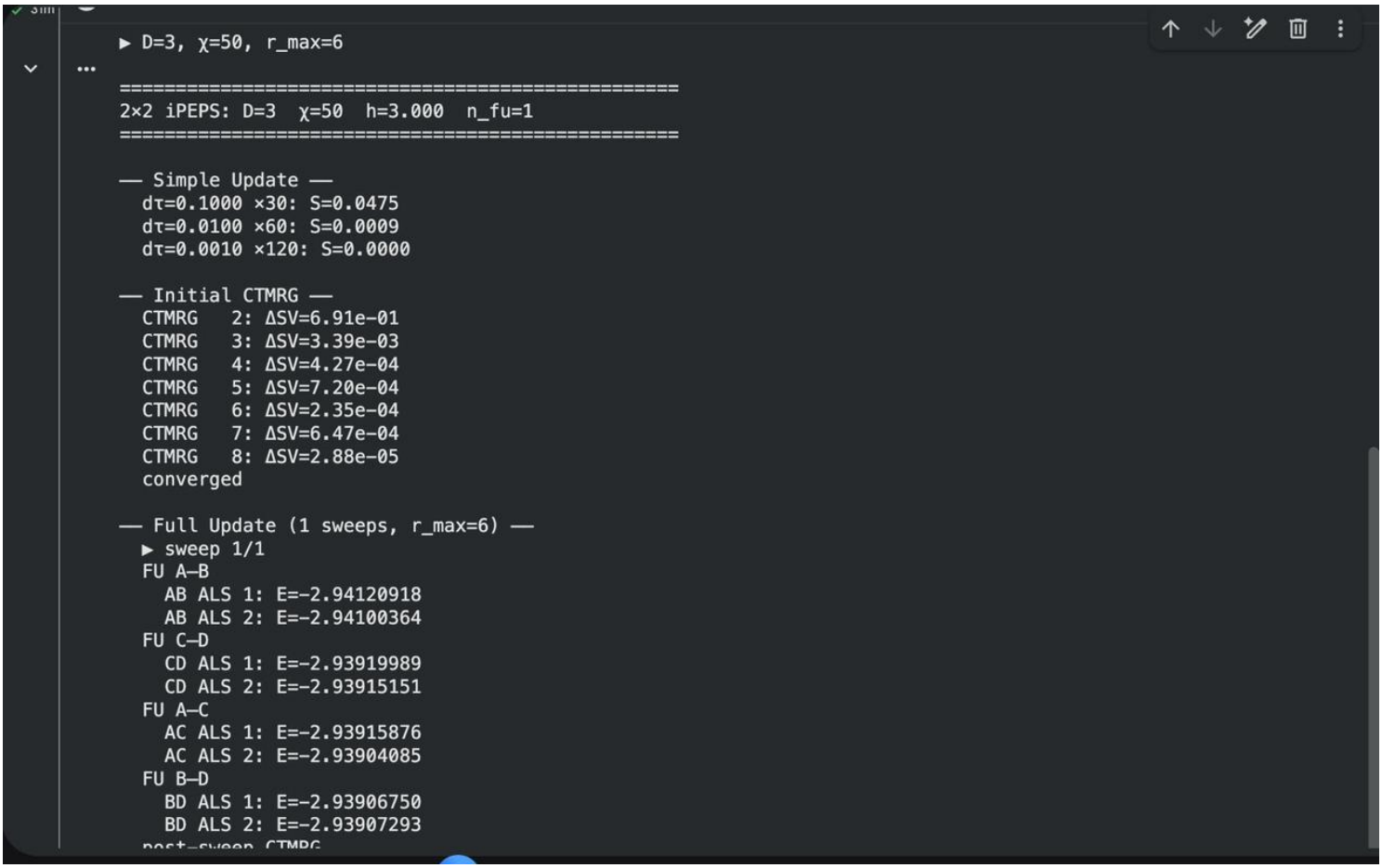
     ================================================
     2×2 iPEPS: D=2  χ=40  h=3.000  n_fu=1
     ================================================

     ── Simple Update ──
       dτ=0.1000 ×30: S=0.0434
       dτ=0.0100 ×60: S=0.0007
       dτ=0.0010 ×120: S=0.0000

     ── Initial CTMRG ──
       CTMRG   2: ΔSV=7.64e-01
       CTMRG   3: ΔSV=1.40e-01
       CTMRG   4: ΔSV=1.45e-04
       CTMRG   5: ΔSV=4.07e-05
       converged

     ── Full Update (1 sweeps, r_max=4) ──
       ▶ sweep 1/1
       FU A–B
         AB ALS 1: E=-2.85708162
         AB ALS 2: E=-2.85753704
       FU C–D
         CD ALS 1: E=-2.85545811
         CD ALS 2: E=-2.85544202
       FU A–C
         AC ALS 1: E=-2.85547861
         AC ALS 2: E=-2.85552229
       FU B–D
         BD ALS 1: E=-2.85563202
```


── full update (1 sweeps, r_max=4) ──
▶ sweep 1/1
FU A–B
AB ALS 1: E=-2.85708162
AB ALS 2: E=-2.85753704
FU C–D
CD ALS 1: E=-2.85545811
CD ALS 2: E=-2.85544202
FU A–C
AC ALS 1: E=-2.85547861
AC ALS 2: E=-2.85552229
FU B–D
BD ALS 1: E=-2.85563392
BD ALS 2: E=-2.85564077
post-sweep CTMRG...
CTMRG 2: ΔSV=4.30e-05
converged
sweep E=-2.8556959587
── Final CTMRG ──
CTMRG 2: ΔSV=1.12e-05
converged
── Observables ──
E/site = -2.8556547729
⟨σᶻ⟩ = -0.486306
⟨σˣ⟩ = 0.872920
S_ent = 0.000009
E_QMC = -3.2802 (reference)
D=2 done in 24 s E=-2.855655
▶ D=3, χ=50, r_max=6
2×2 iPEPS: D=3 χ=50 h=3.000 n_fu=1


▶ D=3, χ=50, r_max=6
2×2 iPEPS: D=3 χ=50 h=3.000 n_fu=1
── Simple Update ──
dτ=0.1000 ×30: S=0.0475
dτ=0.0100 ×60: S=0.0009
dτ=0.0010 ×120: S=0.0000
── Initial CTMRG ──
CTMRG 2: ΔSV=6.91e-01
CTMRG 3: ΔSV=3.39e-03
CTMRG 4: ΔSV=4.27e-04
CTMRG 5: ΔSV=7.20e-04
CTMRG 6: ΔSV=2.35e-04
CTMRG 7: ΔSV=6.47e-04
CTMRG 8: ΔSV=2.88e-05
converged
── Full Update (1 sweeps, r_max=6) ──
▶ sweep 1/1
FU A–B
AB ALS 1: E=-2.94120918
AB ALS 2: E=-2.94100364
FU C–D
CD ALS 1: E=-2.93919989
CD ALS 2: E=-2.93915151
FU A–C
AC ALS 1: E=-2.93915876
AC ALS 2: E=-2.93904085
FU B–D
BD ALS 1: E=-2.93906750
BD ALS 2: E=-2.93907293

```
      BD ALS 1: E=-2.93906750
      BD ALS 2: E=-2.93907293
    post-sweep CTMRG...
    CTMRG   2: ΔSV=8.72e-06
    converged
    sweep E=-2.9391254264

  ── Final CTMRG ──
    CTMRG   2: ΔSV=9.88e-06
    converged

  ── Observables ──
    E/site = -2.9390449152
    ⟨σᶻ⟩   = -0.329117
    ⟨σˣ⟩   = 0.943351
    S_ent  = 0.000012
    E_QMC  = -3.2802  (reference)
    D=3 done in 386 s  E=-2.939045

  ▶ D=4, χ=50, r_max=8

  ================================================
  2×2 iPEPS: D=4  χ=50  h=3.000  n_fu=1
  ================================================

  ── Simple Update ──
    dτ=0.1000 ×30: S=0.0331
    dτ=0.0100 ×60: S=0.0005
    dτ=0.0010 ×120: S=0.0000

  ── Initial CTMRG ──
    CTMRG   2: ΔSV=6.70e-01
    CTMRG   3: ΔSV=2.52e-03
    CTMRG   4: ΔSV=3.72e-04
    CTMRG   5: ΔSV=1.23e-04
```

```
  dτ=0.0100 ×60: S=0.0005
  dτ=0.0010 ×120: S=0.0000
...

── Initial CTMRG ──
  CTMRG   2: ΔSV=6.70e-01
  CTMRG   3: ΔSV=2.52e-03
  CTMRG   4: ΔSV=3.72e-04
  CTMRG   5: ΔSV=1.23e-04
  CTMRG   6: ΔSV=1.42e-04
  CTMRG   7: ΔSV=1.94e-04
  CTMRG   8: ΔSV=2.30e-04
  CTMRG   9: ΔSV=2.34e-04
  CTMRG  10: ΔSV=1.38e-04
  CTMRG  11: ΔSV=2.59e-04
  CTMRG  12: ΔSV=1.46e-04
  CTMRG  13: ΔSV=1.44e-04
  CTMRG  14: ΔSV=2.62e-04
  CTMRG  15: ΔSV=2.85e-04
  CTMRG  16: ΔSV=1.97e-04
  CTMRG  17: ΔSV=1.33e-04
  CTMRG  18: ΔSV=5.94e-04
  CTMRG  19: ΔSV=8.03e-04
  CTMRG  20: ΔSV=2.28e-04
  max_iter reached
  ⚠ FU memory estimate: 3.3e+08 elements (limit 1.0e+08) — SKIP
  ⚠ Skipping Full Update for D=4 — memory limit exceeded
    Continuing with Simple Update + CTMRG only

── Final CTMRG ──
  CTMRG   2: ΔSV=8.94e-04
  converged

── Observables ──
  E/site = -2.7509832926
  ⟨σᶻ⟩   = 0.608228
  ⟨σˣ⟩   = 0.793756
```

```
── Final CTMRG ──
  CTMRG   2: ΔSV=8.94e-04
  converged

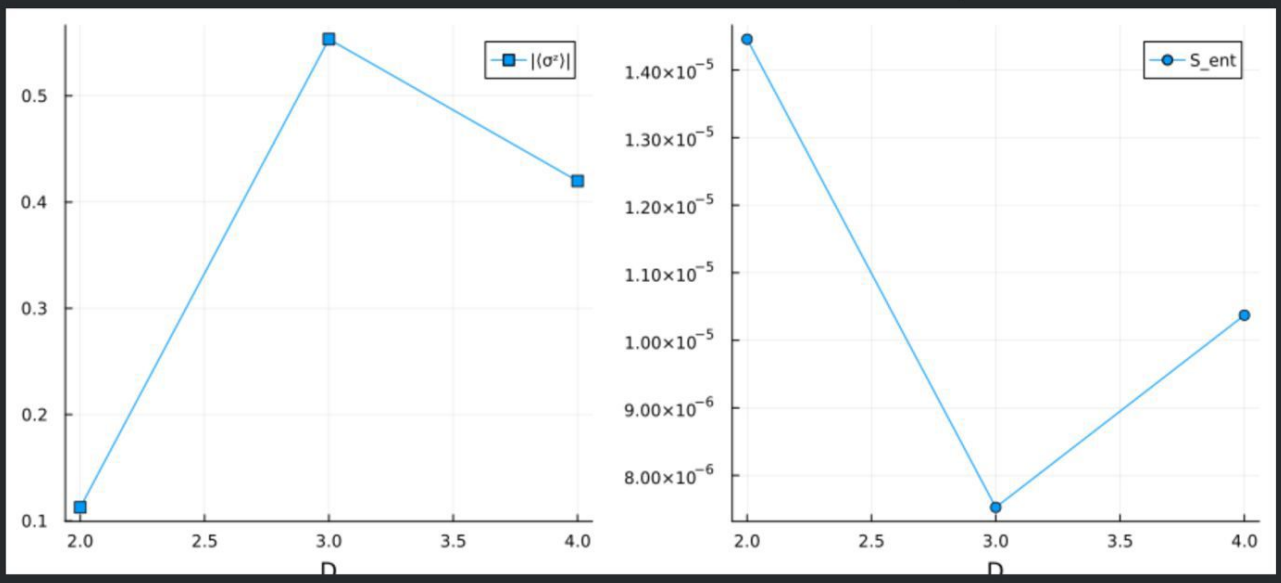
── Observables ──
  E/site = -2.7509832926
  ⟨σᶻ⟩   = 0.608228
  ⟨σˣ⟩   = 0.793756
  S_ent  = 0.000006
  E_QMC  = -3.2802  (reference)
  D=4 done in 1296 s  E=-2.750983

======================================================
SUMMARY
======================================================
D    χ    E/site          S          ⟨σᶻ⟩       ⟨σˣ⟩       Time
------------------------------------------------------------
2    40   -2.85565477     0.000009   -0.486306  0.872920   24
3    50   -2.93904492     0.000012   -0.329117  0.943351   386
4    50   -2.75098329     0.000006   0.608228   0.793756   1296

QMC reference (h=3.0): E/site = -3.2802
Energy should DECREASE with increasing D
3-element Vector{NamedTuple}:
 (D = 2, χ = 40, E = -2.8556547729012363, S = 8.982940957809754e-6, mz = -0.4863064475061636, mx =
0.872920038758739, time = 24.399269821)
 (D = 3, χ = 50, E = -2.939044915243608, S = 1.1977406106692827e-5, mz = -0.32911694872413283, mx =
0.9433507378581842, time = 386.234689803)
 (D = 4, χ = 50, E = -2.750983292582176, S = 6.292514306739058e-6, mz = 0.6082282641331442, mx =
0.7937564371913758, time = 1296.267905305)
```

[8]
✓ 1s

```
plot(layout=(1,2), size=(900,400))
plot!(subplot=1, Ds, [abs(r.mz) for r in results], marker=:square, label="|⟨σᶻ⟩|", xlabel="D")
plot!(subplot=2, Ds, [r.S for r in results], marker=:circle, label="S_ent", xlabel="D")
```



```julia
using Pkg; Pkg.add("Plots"); using Plots
Ds = [r.D for r in results]; Es = [r.E for r in results]
plot(Ds, Es, marker=:circle, markersize=8, lw=2, xlabel="D", ylabel="E/site",
     title="2×2 iPEPS TFIM h=3.0", label="iPEPS")
hline!([-3.2802], ls=:dash, color=:red, label="QMC exact")
```

```
  Resolving package versions...
 No Changes to `~/.julia/environments/v1.11/Project.toml`
 No Changes to `~/.julia/environments/v1.11/Manifest.toml`
```

